GOAL: Create an asynchronous Buck Converter able to take in 10v-15v and output 5v at 1A
*Assume 90% efficiency & tested at 15v,
Must be surface mount parts

Calculations from: https://www.ti.com/lit/an/slva477c/slva477c.pdf?ts=1791641985921
IMPORTANT:
https://www.analog.com/media/en/technical-documentation/data-sheets/13756fd.pdf

Buck IC https://www.digikey.ca/en/products/detail/analog-devices-inc/LT1375CS8-PBF/891548

Based on calcs, inductor:
https://www.digikey.ca/en/products/detail/bourns-inc/SRP1038A-150M/4876660
Diode:
https://www.digikey.ca/en/products/detail/panjit-international-inc/SS2040FL-R1-00001/14660034
Boost:
https://www.digikey.ca/en/products/detail/diodes-incorporated/1N4148W-7-F/814371?s=N4Ig7CBcoIYE5QIwA5EGYA0IYBcmZAAcBLKATgDYBWK5MAX3qA

R1
https://www.digikey.com/en/products/detail/stackpole-electronics-inc/RMCF0603FT5K36/1714091
R2
https://www.digikey.ca/en/products/detail/stackpole-electronics-inc/RMCF0603FT4K99/1761322

1k
https://www.digikey.ca/en/products/detail/stackpole-electronics-inc/RMCF0603FT1K00/1761077

0.05 https://www.digikey.ca/en/products/detail/vishay-dale/WSL0603R0500FEA18/1882601
input cap https://www.digikey.ca/en/products/detail/kemet/C1210C226M3RACTU/3317484

Capacitor Boot https://www.digikey.ca/en/products/detail/kemet/C0603C104K5RACTU/1465594

Output Capacitor:
https://www.digikey.ca/en/products/detail/kemet/C1210C107M4PACTU/4437452

3.3 nF cap https://www.digikey.com/en/products/detail/kemet/C0603C332K5RACTU/411085

Simulate: https://www.analog.com/en/products/lt1375.html#overview

Calculations

$V_{IN(max)} = 15V$

$V_{OUT} = 5V$

* assume 90% efficiency

$$D = \frac{V_{OUT}}{V_{IN(max)} \times \eta} = \frac{5V}{(15V)0.9} = 0.37 = \boxed{37\%}$$

Inductor value: 3μH – 20μH

$\therefore L = 18\mu H$

Inductor Ripple Current:

$$\Delta I_L = \frac{(V_{IN(max)} - V_{OUT}) \times D}{f_s \times L}$$

$$= \frac{(15V - 5V) \times 37\%}{(440kHz)(18\mu H)}$$

$$= 0.468A$$

$f_s = 440\ kHz$

$DC \leq 50\% \Rightarrow I_{Lim} = 1.50A$

$$I_{Maxout} = I_{Lim} - \frac{\Delta I_L}{2}$$

$$= 1.50A - \frac{0.468A}{2}$$

$$= 1.266A$$ → amount IC supports

$$I_{SW(max)} = I_{out(max)} + \frac{\Delta I_L}{2}$$

$$= 1A + \frac{0.468A}{2}$$

$$= 1.234A$$ → peak switch current

Diode:

$$I_F = I_{OUT(max)} \times (1 - D)$$

$$= 1A \times (1 - 0.37)$$

$$= 0.63A$$

$$P_D = I_F \times V_F$$

$$= 0.63A \times (0.5V)$$

$$= 0.315W$$

Resistors:

on data sheet

Input Capacitor:

on sheet

Output Capacitor:

on sheet



LTSpice Logs:
V1: vavg: AVG(V(vout))=5.01524832439 FROM 0.008 TO 0.01
vmax: MAX(V(vout))=5.03797674179 FROM 0.008 TO 0.01
vmin: MIN(V(vout))=4.99225521088 FROM 0.008 TO 0.01
ripple: VMAX-VMIN=0.0457215309143

V2: vavg: AVG(V(vout))=5.01524448424 FROM 0.008 TO 0.01
vmax: MAX(V(vout))=5.02699804306 FROM 0.008 TO 0.01
vmin: MIN(V(vout))=5.00335931778 FROM 0.008 TO 0.01
ripple: VMAX-VMIN=0.0236387252808

Reference: vavg: AVG(V(vout))=5.01499302422 FROM 0.008 TO 0.01
vmax: MAX(V(vout))=5.03242349625 FROM 0.008 TO 0.01
vmin: MIN(V(vout))=4.99728536606 FROM 0.008 TO 0.01
ripple: VMAX-VMIN=0.035138130188
Untitled spreadsheet

Design	Ripple (mV)
LT1375 Reference	35.138
v1	45.722
v2	23.639

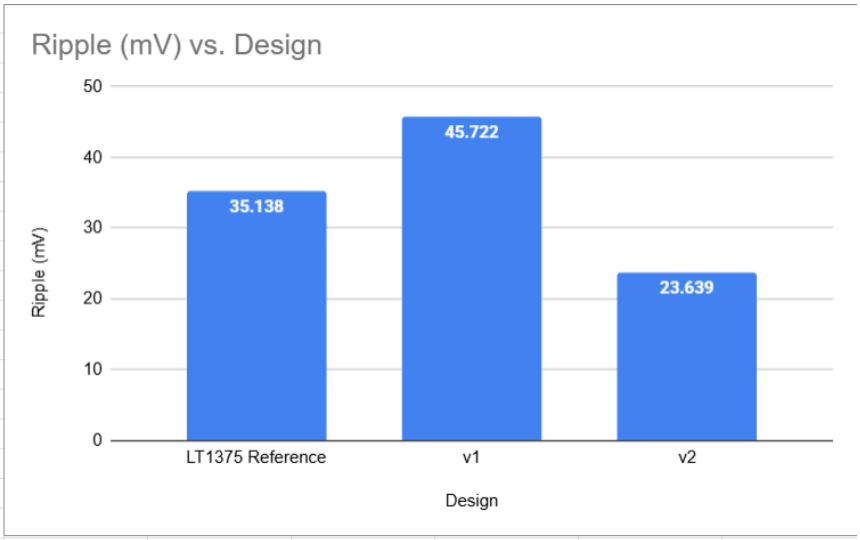
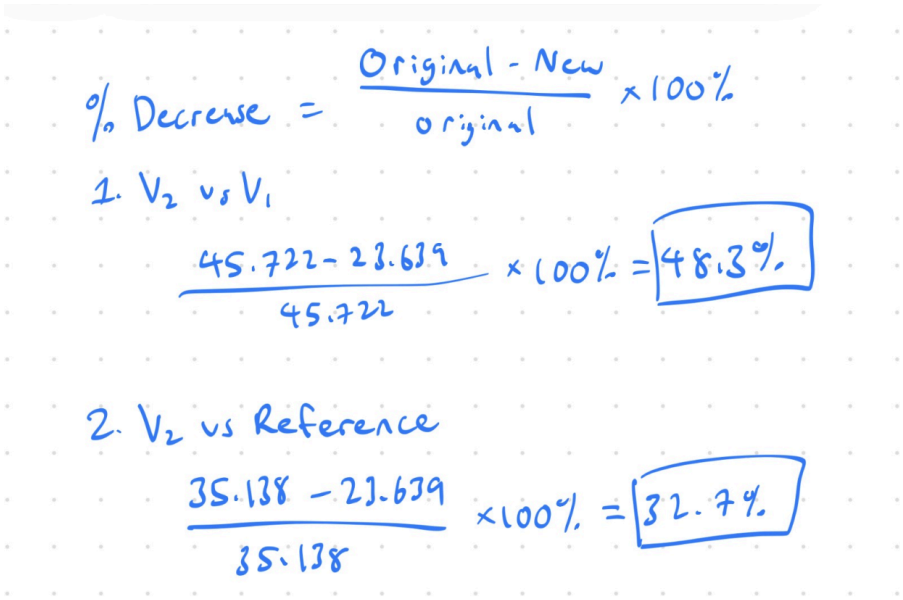


Figure 1. Comparison of simulated peak-to-peak output voltage ripple for the LT1375 reference circuit, initial design (V1), and optimized design (V2). Measurements were taken from 8–10 ms at 15 V input and a 5 Ω load.

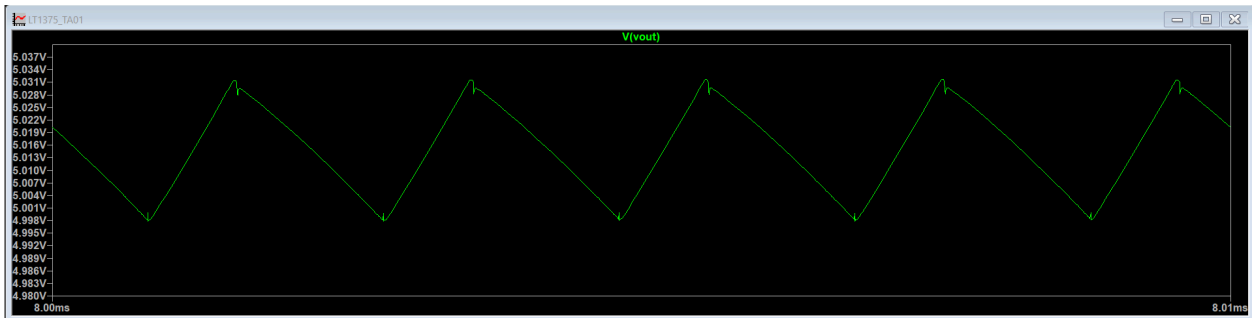


The optimized buck converter (V2) achieved 23.639 mV peak-to-peak output ripple, compared with 45.722 mV for V1 and 35.138 mV for the manufacturer's reference circuit. This represents a 48.3% reduction in simulated ripple compared with V1 and a **32.7%** reduction compared with the reference design under the same test conditions.

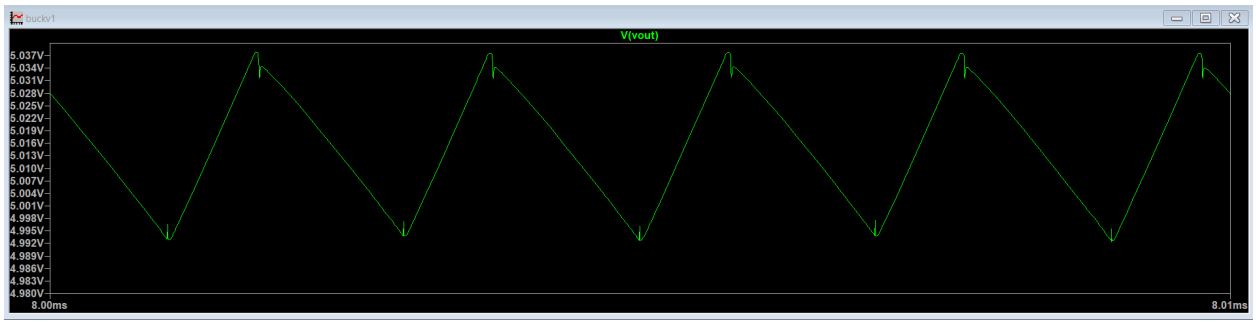
What I did:
V1 used a 15 uH inductor, a 100 uF output capacitor with 0.10 ohm modeled ESR, and a 3.3 nF VC compensation capacitor.

For V2, I decreased R4 from 0.1 Ω to 0.05 Ω to reduce the output capacitor's ESR. Since the capacitor handles changing current when the buck converter switches, having less resistance means less voltage fluctuation and therefore lower ripple. I also added a 1 kΩ resistor to the compensation circuit to help control how the IC responds to voltage changes and prevent oscillations. After making these changes, the simulated output ripple decreased from 45.722 mV to 23.639 mV, which is a 48.3% improvement over V1.

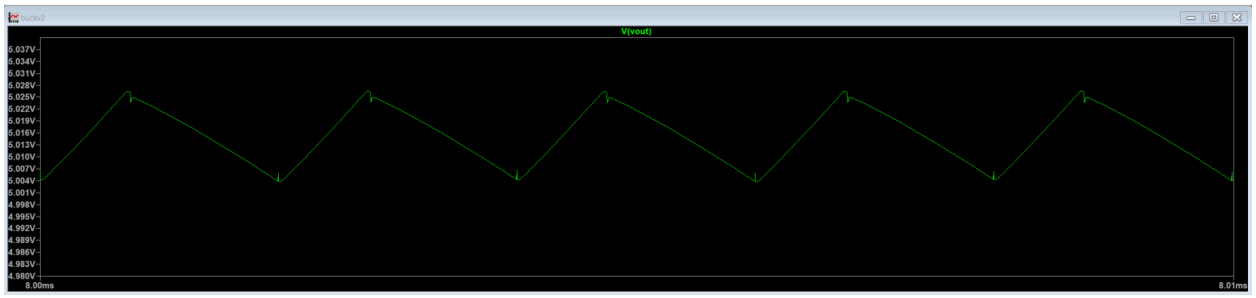
Reference Vout



v1 Vout



v2 Vout



Altium Traces:
https://www.digikey.ca/en/resources/conversion-calculators/conversion-calculator-pcb-trace-width

CURRENT (I)

1.5 A

THICKNESS (T)

1 oz/ft²

TEMPERATURE RISE (T_{RISE})

10 °C

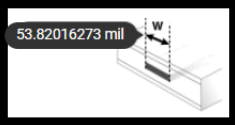
AMBIENT TEMPERATURE

40 °C

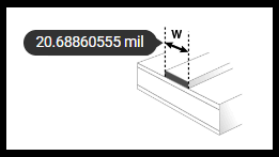
TRACE LENGTH

1 in

Minimum Trace Width



Minimum Trace Width



Internal Layers

REQUIRED TRACE WIDTH (W)

53.82016273 mil

RESISTANCE

0.009904673522 Ω

VOLTAGE DROP

0.01485701028 V

POWER LOSS

0.02228551542 W

External Layers in Air

REQUIRED TRACE WIDTH (W)

20.68860555 mil

RESISTANCE

0.02576641231 Ω

VOLTAGE DROP

0.03864961846 V

POWER LOSS

0.05797442770 W